Exact One-Measurement Recovery and a Sharper General Upper Bound for Noisy Diagonal Linear Information

Automated mathematics run `fa_banach_001`, lane 3

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Abstract

Krieg, Novak, Plaskota, and Ullrich ask for the exact worst-case error of recovering a diagonal image of the unit ball of ℓ_p from norm-one linear measurements corrupted by coordinatewise adversarial noise. For every $1 \le p < \infty$, we solve the one-measurement case exactly:

$$e_1^{\mathrm{lin}}(D_\sigma, \delta)_p = \left(\delta^p \sigma_1^p + (1-\delta^p)\sigma_2^p\right)^{1/p}.$$

For arbitrary n, we also construct a nonlinear decoder that strictly improves the coordinate decoder displayed in the source. It gives

$$e_n^{\mathrm{lin}}(D_\sigma, \delta)_p^p \le \min_{0 \le k \le n} \left\{ \sigma_{k+1}^p + \delta^p \sum_{i=1}^{k} (\sigma_i^p - \sigma_{k+1}^p) \right\}.$$

The matching lower bound for $n \ge 2$ remains open. Status: *candidate partial result, likely valid, pending human review.*

1 Source question

The source is D. Krieg, E. Novak, L. Plaskota, and M. Ullrich, *Noisy nonlinear information and entropy numbers*, arXiv:2510.23213 (2025), subsequently published in 2026 [1]. Section 5.2 of the published version, and PDF page 12 of the arXiv version, state the question shown in Figure 1.

$\widetilde{A}_n(x) = (\sigma_1 y_1, \sigma_2 y_2, \ldots, \sigma_n y_n, 0, 0, 0 \ldots)$, where $|y_i - x_i| \le \delta$, $1 \le i \le n$. Then for $1 \le p < \infty$ we have

$$e(\widetilde{A}_n, \delta)_p = \left(\sup_{\|x\|_p \le 1} \sup_{|y_i - x_i| \le \delta} \sum_{i=1}^{n} \sigma_i^p |y_i - x_i|^p + \sum_{j=n+1}^{\infty} \sigma_j^p |x_j|^p \right)^{1/p}$$

$$= \left(\delta^p \sum_{i=1}^{n} \sigma_i^p + \sigma_{n+1}^p \right)^{1/p}, \tag{7}$$

while for $p = \infty$ we have $e(\widetilde{A}_n, \delta)_\infty = \max(\delta\sigma_1, \sigma_{n+1})$.

For a lower bound we have that for any p the error of any approximation using n measurements is at least $\max(\delta\sigma_1, \sigma_{n+1})$. Indeed, since for any x with $\|x\|_p \le \delta$ and any λ it holds that $|\lambda x| \le \delta$, all elements of the ball of radius δ and center at 0 are indistiguishable with respect to zero information. Hence, on one hand, the error of any approximation is lower bounded by

$$\sup\{\|D_\sigma x\|_p \mid \|x\|_p \le \delta\} = \delta\sigma_1,$$

and, on the other hand, it is not smaller than the minimal error σ_{n+1} from exact measurements. This means that

$$e_n^{\mathrm{lin}}(D_\sigma, \delta)_\infty = \max(\delta\sigma_1, \sigma_{n+1}).$$

Unfortunately, the approximation $\widetilde{A}_n$ is nth optimal only for $p = \infty$. What is the exact value of $e_n^{\mathrm{lin}}(D_\sigma, \delta)_p$ for $p < \infty$ is an open problem.

Remark 10. Suppose $p = 2$. Then the nth minimal error is known exactly in very special cases only. One of these is when $\sigma_{n+1} = 0$ and the worst case error is taken over the whole space ℓ_2 (instead of the unit ball). Then (7) is valid and the nth minimal error equals $\delta\sqrt{\sum_{i=1}^n \sigma_i^2}$, see the Appendix of [15].

It is also worthwhile to mention that much more is known if we assume that the noise is bounded in the ℓ_2-norm (instead of ℓ_∞), i.e., $\sum_{i=1}^n |y_i - \lambda_i x|^2 \le \delta^2$. Then the corresponding nth minimal error equals

$$\widehat{e}_n^{\mathrm{lin}}(D_\sigma, \delta)_2 = \sqrt{\sigma_{n+1}^2 + \frac{\delta^2}{n} \sum_{j=1}^{n} (\sigma_j^2 - \sigma_{n+1}^2)},$$

see [14, p.79]. Obviously $\widehat{e}_n^{\mathrm{lin}}(D_\sigma, \delta)_2 \le e_n^{\mathrm{lin}}(D_\sigma, \delta)_2 \le \widehat{e}_n^{\mathrm{lin}}(D_\sigma, \delta\sqrt{n})_2$.

Figure 1: The source upper bound, lower bound, and open question, cropped from arXiv PDF page 12.

Let $1 \le p < \infty$, let B_p be the unit ball of ℓ_p, and let $\sigma_1 \ge \sigma_2 \ge \cdots \ge 0$. The diagonal map is

$$D_\sigma(x_1, x_2, \ldots) = (\sigma_1 x_1, \sigma_2 x_2, \ldots).$$

An algorithm may choose n linear functionals of dual norm at most one (adaptively after the first), observe numbers y_i satisfying $|y_i - \lambda_i(x)| \le \delta$, and apply an arbitrary reconstruction map. Its worst-case error is measured in ℓ_p, and the infimum of that error is $e_n^{\mathrm{lin}}(D_\sigma, \delta)_p$. The source proves

$$\max\{\delta\sigma_1, \sigma_{n+1}\} \le e_n^{\mathrm{lin}}(D_\sigma, \delta)_p \le \left(\delta^p \sum_{i=1}^{n} \sigma_i^p + \sigma_{n+1}^p\right)^{1/p}, \tag{1}$$

and asks for the exact value when $p < \infty$.

2 Proof intuition

The source decoder returns the noisy coordinate y_i itself. This treats measurement error and the entire unobserved tail as though both could be maximal at once, ignoring the unit-ball constraint. The improvement is to shrink each y_i toward zero by an amount chosen from the whole feasible interval $[y_i - \delta, y_i + \delta] \cap [-1, 1]$. A one-dimensional interval intersection lemma chooses a center that exactly trades coordinate error against the remaining ℓ_p mass.

For one measurement, this decoder attains the stated formula. The reverse inequality needs no assumption about the chosen functional: in the first two coordinate directions, a connected cap of the ℓ_p unit sphere contains a point with $|x_1| \ge \delta$ whose measurement is compatible with the zero datum. The two antipodal inputs x and $-x$ are then indistinguishable.

For several measurements, the scalar centers can be chosen independently. This yields the sharper upper bound, but the cap argument does not simultaneously control n arbitrary slabs. That missing noisy Gelfand-width lower bound is precisely why the original problem remains open for $n \ge 2$.

3 The scalar-center lemma

Lemma 3.1. *Fix $1 \le p < \infty$, $0 \le c \le 1$, and $0 < \delta < 1$. Define*

$$h(u) = \big((1-c)\delta^p + c|u|^p\big)^{1/p}, \qquad -1 \le u \le 1.$$

For every nonempty interval $I = [-1, 1] \cap [y - \delta, y + \delta]$, there is a real number $t = t(y)$ such that

$$|u - t|^p \le (1-c)\delta^p + c|u|^p \qquad (u \in I). \tag{2}$$

Proof. The function h is even and convex: it is the ℓ_p^2 norm of $(c^{1/p}u, (1-c)^{1/p}\delta)$. Moreover, $u - h(u)$ and $u + h(u)$ are nondecreasing. For $p > 1$ this follows from

$$|h'(u)| = \frac{c|u|^{p-1}}{h(u)^{p-1}} \le c^{1/p} \le 1,$$

and for $p = 1$ it is immediate from the two linear pieces.

Write $I = [l, r]$ and $d = r - l \le 2\delta$. Since h is even and convex, the sum $h(s - d/2) + h(s + d/2)$ is an even convex function of s, and hence

$$h(l) + h(r) \ge 2h(d/2) \ge d.$$

Thus $r - h(r) \le l + h(l)$. Monotonicity shows

$$\max_{u\in I}(u - h(u)) = r - h(r) \le l + h(l) = \min_{u\in I}(u + h(u)).$$

Consequently the intervals $[u - h(u), u + h(u)]$, $u \in I$, have a common point. Any such point is t and gives (2). For example, one may take the midpoint of $[r - h(r), l + h(l)]$. $\square$

4 A sharper upper bound for every number of measurements

Theorem 4.1. *For $1 \le p < \infty$, $0 < \delta < 1$, and every integer $n \ge 0$,*

$$e_n^{\mathrm{lin}}(D_\sigma, \delta)_p^p \le \min_{0\le k\le n} U_k, \qquad U_k := \sigma_{k+1}^p + \delta^p \sum_{i=1}^{k} (\sigma_i^p - \sigma_{k+1}^p). \tag{3}$$

In particular, using $k = n$ improves the pth power of the source upper bound in (1) by $n\delta^p\sigma_{n+1}^p$.

Proof. Fix $k \le n$ and measure $x_1, \ldots, x_k$; any remaining measurements are ignored. Put $b = \sigma_{k+1}^p$. Given a consistent noisy value y_i, apply Lemma 3.1 to its feasible interval with $c_i = b/\sigma_i^p$ when $\sigma_i > 0$, obtaining $t_i = t_i(y_i)$. If $\sigma_i = 0$, set $t_i = 0$. Reconstruct

$$A(y) = (\sigma_1 t_1, \ldots, \sigma_k t_k, 0, 0, \ldots).$$

For every consistent input $x \in B_p$, Lemma 3.1 gives

$$\sigma_i^p |x_i - t_i|^p \le (\sigma_i^p - b)\delta^p + b|x_i|^p.$$

Since $\sigma_j^p \le b$ for $j > k$,

$$\begin{aligned}\|D_\sigma x - A(y)\|_p^p &\le \sum_{i=1}^{k} \big((\sigma_i^p - b)\delta^p + b|x_i|^p\big) + b\sum_{j>k} |x_j|^p \\ &\le b + \delta^p \sum_{i=1}^{k} (\sigma_i^p - b) = U_k.\end{aligned}$$

Taking the best k proves (3). For $k = n$, expanding U_n gives $\delta^p \sum_{i=1}^{n} \sigma_i^p + (1 - n\delta^p)\sigma_{n+1}^p$, which is the source expression minus $n\delta^p\sigma_{n+1}^p$. $\square$

Remark 4.2. *The minimum over k matters when $k\delta^p > 1$. Consecutive values satisfy*

$$U_{k+1} - U_k = (1 - k\delta^p)(\sigma_{k+2}^p - \sigma_{k+1}^p),$$

so the displayed family decreases until roughly $k = \delta^{-p}$ and can increase afterward. Extra measurements can always be ignored.

5 Exact solution for one measurement

Theorem 5.1. *For every $1 \le p < \infty$ and $0 < \delta < 1$,*

$$\boxed{e_1^{\mathrm{lin}}(D_\sigma, \delta)_p = \big(\delta^p \sigma_1^p + (1 - \delta^p)\sigma_2^p\big)^{1/p}.} \tag{4}$$

The formula also extends to $\delta = 0$.

Proof. The upper bound is Theorem 4.1 with $k = 1$.

For the lower bound, let λ be an arbitrary admissible first functional. Restrict it to $E = \mathrm{span}\{e_1, e_2\}$ and write $\alpha = \lambda(e_1)$ and $\beta = \lambda(e_2)$. We claim that there is an $x \in E$ with

$$\|x\|_p = 1, \qquad |x_1| \geq \delta, \qquad |\lambda(x)| \leq \delta. \tag{5}$$

If $|\alpha| \leq \delta$, take $x = e_1$. Otherwise, changing signs if necessary, assume $\alpha > \delta$. Let $s = 1$ if $\beta \geq 0$ and $s = -1$ otherwise, and let

$$z = \delta e_1 - s(1 - \delta^p)^{1/p} e_2.$$

Then $\|z\|_p = 1$ and $\lambda(z) = \alpha\delta - |\beta|(1-\delta^p)^{1/p} \leq \alpha\delta \leq \delta$, because $|\alpha| \leq \|\lambda\| \leq 1$. Along the connected unit-sphere arc from e_1 to z on which the first coordinate decreases from 1 to δ, continuity gives a point with $\lambda(x) = \delta$. This proves (5).

Both x and $-x$ are compatible with the same datum $y = 0$. For any reconstruction a at that datum, the triangle inequality gives

$$\max\{\|D_\sigma x - a\|_p, \|-D_\sigma x - a\|_p\} \geq \|D_\sigma x\|_p.$$

Since x has only its first two coordinates nonzero,

$$\begin{aligned} \|D_\sigma x\|_p^p &= \sigma_1^p |x_1|^p + \sigma_2^p (1 - |x_1|^p) \\ &\geq \delta^p \sigma_1^p + (1 - \delta^p)\sigma_2^p. \end{aligned}$$

This lower bound holds for every admissible λ and every reconstruction, proving (4). ☐

Corollary 5.2. *For rank-one D_σ (that is, $\sigma_2 = 0$), $e_n^{\mathrm{lin}}(D_\sigma, \delta)_p = \delta\sigma_1$ for every $n \geq 1$. If $\sigma_1 = \cdots = \sigma_{n+1}$, then $e_n^{\mathrm{lin}}(D_\sigma, \delta)_p = \sigma_1$.*

Proof. In the rank-one case Theorem 5.1 supplies an algorithm with one measurement and the source lower bound supplies the reverse inequality for every n. In the flat-spectrum case the exact-information lower bound σ_{n+1} matches the zero-output upper bound. ☐

6 Verification, scope, and novelty status

The script `code/verify_scalar_center.py` checks the endpoint intersection and decoder inequality on a deterministic grid and random instances. It is a sanity check only; the proof is the analytic argument above. A separate Euclidean random-frame probe in the run's attempt record found no smaller zero-data radius than coordinate information for $n = 2$ in four tested three-dimensional spectra, but this is not used in any claim.

The full exact-value problem remains open for $n \geq 2$. In particular, no matching lower bound for (3) is claimed. Such a result would have to control the intersection of the ℓ_p ball with n arbitrary adaptive zero-data slabs; the one-cap intermediate-value argument does not provide that simultaneous control.

A bounded novelty search on 11 August 2026 used the exact title, arXiv id, the exact displayed notation, and the phrases "noisy linear information" and "diagonal operator". It checked the arXiv record and the revised Journal of Fourier Analysis and Applications article, published 16 March 2026. The revised article still states the question as open in Section 5.2, and no later paper settling this one-measurement formula or the sharper upper bound was found. This is not an exhaustive literature review.

Human review is recommended, focusing on the interval-intersection step in Lemma 3.1, the admissibility of the data-dependent scalar centers, and the cap-continuity lower bound in Theorem 5.1.

References

[1] D. Krieg, E. Novak, L. Plaskota, and M. Ullrich, *Noisy nonlinear information and entropy numbers*, J. Fourier Anal. Appl. **32**, Article 36 (2026), doi:10.1007/s00041-026-10249-z; arXiv:2510.23213.